Lecture 3(a): Flexibility definitions, stakeholders, models

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Agenda

- Flexibility definitions
- Stakeholders
- Ownership
- Matching grid needs with available flexibility

Slido questions

- Do you know the role of a system operator?
  - Y/N
- Do you know the role of an aggregator?
  - Y/N

Do you know the role of a system operator?

Yes

0%

No

0%

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Do you know the role of an aggregator?

Yes

0%

No

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Flexibility definitions

Existing flexibility definitions

- *ENTSO-e definition for flexibility*: Power system flexibility is defined as the ability of a power system to effectively manage the **variability and uncertainty of electricity demand, supply, and grid availability** across a range of timeframes, from seconds to years

- *Eurelectric definition*: Flexibility, i.e. the ability of the power system to **adapt to fluctuations of supply, demand, or grid availability**, is becoming a cornerstone of security of supply in this evolving landscape

- *IEA definition*: the ability of a power system to **reliably and cost-effectively manage the variability and uncertainty** of supply and demand across all relevant timescales

Ref1: "ENTSO-E Vision: A Power System for a Carbon Neutral Europe — vision.entsoe.eu," https://vision.entsoe.eu/
Ref2: "eurelectric.org," https://www.eurelectric.org/wp-content/uploads/2025/02/Eurelectric-report-energy-security-in-the-age-of-electricity.pdf
Ref3: IEA, "Status of Power System Transformation 2019," https://www.iea.org/reports/status-of-power-system-transformation-2019

Existing definitions

CIRED definition: Flexibility is a power modulation of any flexible resources in voluntary response to a need (a signal). This response allows the power system operator or other third parties to optimize their operation condition (e.g. costs, voltage profile) without affecting their security and reliability. It can be specified that:

- The power modulated can be either active, reactive, or both;
- The modulation consists of an increase or a decrease in power output/input;
- The resources can be a customer through the control of its devices (heaters, electric vehicles, etc.), industrial customers, a production, a storage system, an energy community, or a microgrid;
- Assets owned by the DSO can be seen as a source of flexibility in some countries;
- The signal can be direct (control action) or indirect (incentives or restrictions on use);
- Services such as congestion management, peak management, postponing or avoiding grid reinforcement, frequency service (in real-time), schedule balancing support, voltage support, synthetic inertia, power loss reduction, optimization of self-consumption, phase balancing, increasing the hosting capacity of DERs

Source:
https://www.researchgate.net/publication/400081500_Cired_Working_Group_Report_Flexibility_in_distribution_grids_with_a_large_share_of_distributed_energy_resources

Existing definitions

- EDSO definition: “Flexibility is defined as the modification of generation injection and/or consumption patterns, on an individual or aggregated level, often in reaction to an external signal, in order to provide a service within the energy system or maintain stable grid operation.”

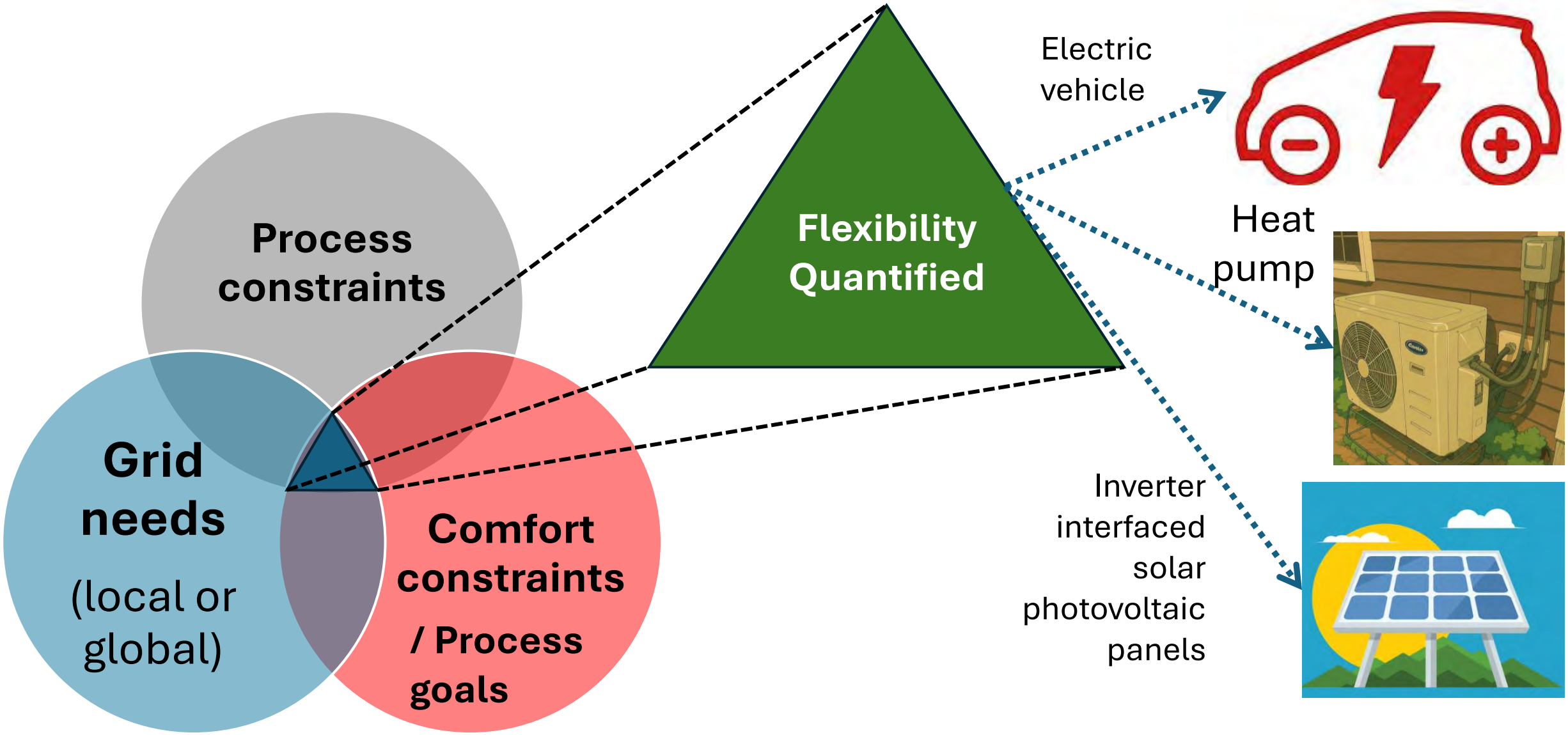
- EU definition: The Electricity Regulation (EU) 2019/943 defines flexibility as the ability of the electricity system to adjust to the variability of generation and consumption patterns and to the grid availability, across relevant market timeframes

Source:
https://cedec.com/files/default/flexibility-in-the-energy-transition-a-tool-for-electricity-dsos-2018.pdf
https://eur-lex.europa.eu/eli/reg/2024/1747/oj

Proposed flexibility definition

- Flexibility is the technical ability of a power system unit to modulate its power feed-in or power out-feed over time (for both active and reactive power) considering local objectives (comfort, QoS, deadline constraint etc.) of the flexibility providing asset while considering local or global grid needs (either based on preset rule or real-time activation signal generated by the market or system operator) across relevant timescales.

Proposed flexibility definition



Flexibility concerns for stakeholders

Stakeholders	Grid needs	Process constraints	Comfort constraints	Aggregated resource adequacy	Billing	Locational and opprtunity Fairness
DSO	✓			✓	✓	✓
Aggregator	✓	✓	✓	✓	✓	✓
Market operator	✓			✓	✓	✓
Flexibility owner		✓	✓		✓	✓
Passive consumers					✓	✓
Active consumers		✓	✓		✓	✓
Policy makers	✓	✓	✓	✓	✓	✓

Flexibility stakeholders

Power network stakeholders

*non exhaustive list

Distribution network stakeholders

*non exhaustive list

Who is a transmission/distribution **system operator**?

A System Operator (TSO/DSO) is the entity responsible for operating, maintaining, and developing the electricity grid.

Main responsibilities:

(a) Real-time system operation: Congestion mitigation and stability management, Grid service procurement

(b) Network planning and development

(c) Validate/approve incoming connection requests for DERs

(d) Smart metering and data management

(e) Ensuring fair opportunity to all suppliers, aggregators, prosumers

How do they make money:→ Regulated tariffs not electricity sales

Network tariffs covers network operation, maintenance, new investments, cost of grid services

Connection fees, feasibility study fees, smart meter fees

Both TSOs and DSOs earn money through regulated tariffs, not electricity sales. Profits are based on a regulator-approved return on the grid assets they operate.

Example: Engie, Fluvius, RTE, Elia, ...

Who is a transmission/distribution **aggregator**?

Key mediator between consumers/producers, markets and other power system participants

Main responsibilities:

Aggregate/bundles the flexible capacity of many small DERs or consumers—such as EVs, batteries, heat pumps, PV inverters, and industrial loads— (act as a large virtual power plant) so that their combined flexibility can participate in energy and grid service markets that individual devices cannot access alone

- interface between consumers/prosumers and the energy markets or system operators
- For consumers/flex assets: (i) Collect & manage flexibility from many devices (EVs, PV, BESS, HVAC). (ii) Optimize behind-the-meter assets to reduce bills or generate revenue. (iii) Respect comfort constraints (e.g., indoor temperature limits). (iv) Provide forecasting of load, generation, and flexibility availability.

Provides grid services to power system stakeholders (TSO, DSO) via market products

Manages the risks associated with uncertainties in the markets and responsiveness of the distributed flexible asset base

- Using consumption and flexibility forecasting
- Market and consumer portfolio management; considers comfort, minimum volume, location and time constraints

Maximize value of consumer's flexibility

Interacts with consumers through price and volume signals assesses their response and behavior

How do they make money: Market participation: Energy arbitrage for market signals

Examples: Powernaut, Next Kraftwerke, tiko Energy Solutions (Switzerland, EU), Voltalis (France), Sonnen (Germany), OhmConnect (USA), Flexitricity (UK)

Who is a market **operator**?

Market Operator (MO) is the entity responsible for organizing, administering, and managing electricity markets, ensuring transparent, competitive, and non-discriminatory trading of electricity and flexibility services.

MO **do not own assets** and **do not operate the grid**

Main responsibilities:

- Operating energy markets
- Market clearing and price formation
- Settlement and financial transactions
- Participant registration and compliance

How do they make money:

- Trading fees: charged per volume of traded energy
- Charges associated with data services
- If TSO acts as MO, revenue driven by regulated tariffs

Example: EPEX SPOT, Nord Pool, ...

Who is a **retailer**?

Retailer: company that buys electricity from wholesale markets and sells it to end-consumers (households, SMEs, industries). Retailers do not own transmission or distribution networks—TSOs and DSOs do.

Main responsibilities:

Retailers buy electricity from: (i) Day-Ahead markets, (ii) Intraday markets, (iii) Bilateral contracts (PPAs), (iv) Long-term hedges or forward markets, (v) Local flexibility markets(in emerging models).They forecast customer demand and procure energy accordingly.

Consumer related responsibilities:

- Pricing and billing: set tariff structures
- Customer service

System level responsibilities:

- Balancing responsibility
- Data handling
- Supporting DER integration

How do they make money:

Trading fees: charged per volume of traded energy

Charges associated with data services

If TSO acts as MO, revenue driven by regulated tariffs

Example: ENGIE, EDF, Octopus Energy, British Gas, Iberdrola, E.ON, EDF, Luminus

Who is an energy policymakers and regulators?

Energy policymakers are government bodies that set national or regional energy strategy, laws, and long-term policy directions. They do not run electricity networks or markets directly. Instead, they define what the energy transition should achieve.

A **regulator** is an independent authority that enforces energy rules, oversees market actors, and protects consumers. Regulators ensure that: (i) electricity markets are fair and transparent, (ii) utilities (TSOs, DSOs, retailers) behave neutrally, (iii) tariffs are reasonable, (iv) grid operators don't exploit monopoly power. They are usually separate from policymakers to avoid political interference.

How do they make money:

Policymakers are government bodies, not commercial entities.

On the other hand, regulators operate as non-profit public authorities.

Example policymakers: EU: European Commission, ACER (policy advisory)UK: Department for Energy Security & Net Zero (DESNZ), US: Department of Energy (DOE). **Example regulators**: Federal Energy Regulatory Commission (FERC) – US, Ofgem – UK, CREG – Belgium, ACER – EU (coordinates national regulators)

Who is a **flexible asset owner**?

A flexible asset owner is any individual, business, or organization that owns a device capable of adjusting electricity consumption, production, or storage in response to external signals (price, market demand, grid needs).

Main responsibilities:

- Complying with connection agreement drafted by system operator
- Ensuring local comfort goals associated with the asset

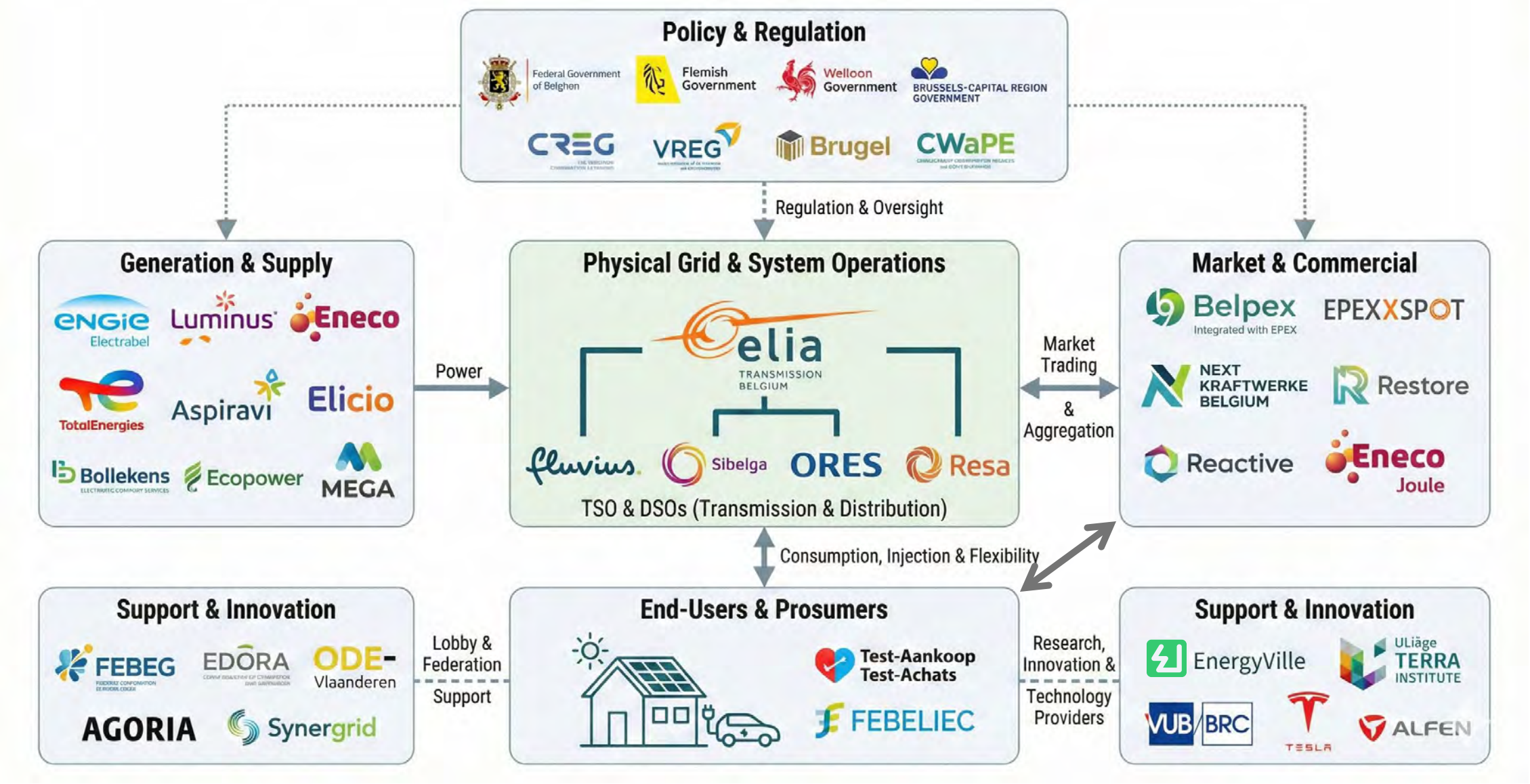
How do they make money:

- Direct market participation
- Payments from aggregators

Example: Batteries (home / commercial), Electric vehicles, Heat pumps, HVAC systems, Flexible industrial loads, Smart home appliances, Community or commercial flexibility assets (e.g., community storage).

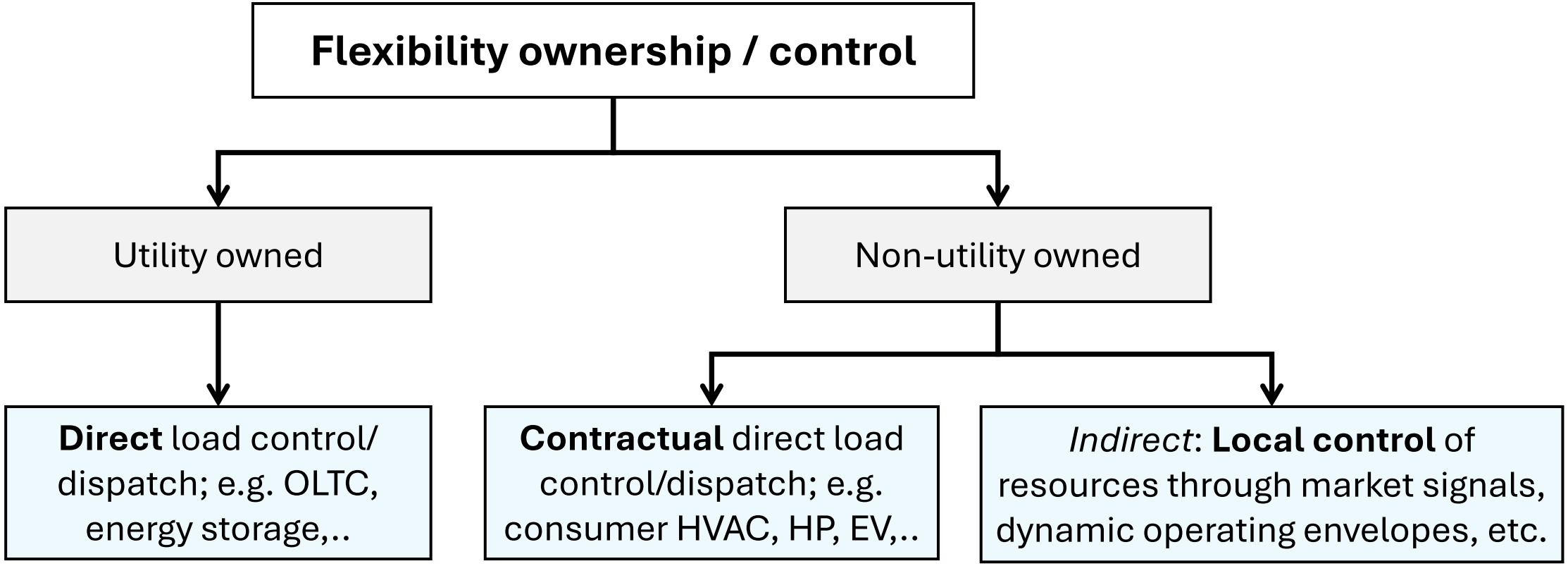
The Belgian Electricity Market Ecosystem

Stakeholder Interaction & Physical Flows



Flexibility ownership

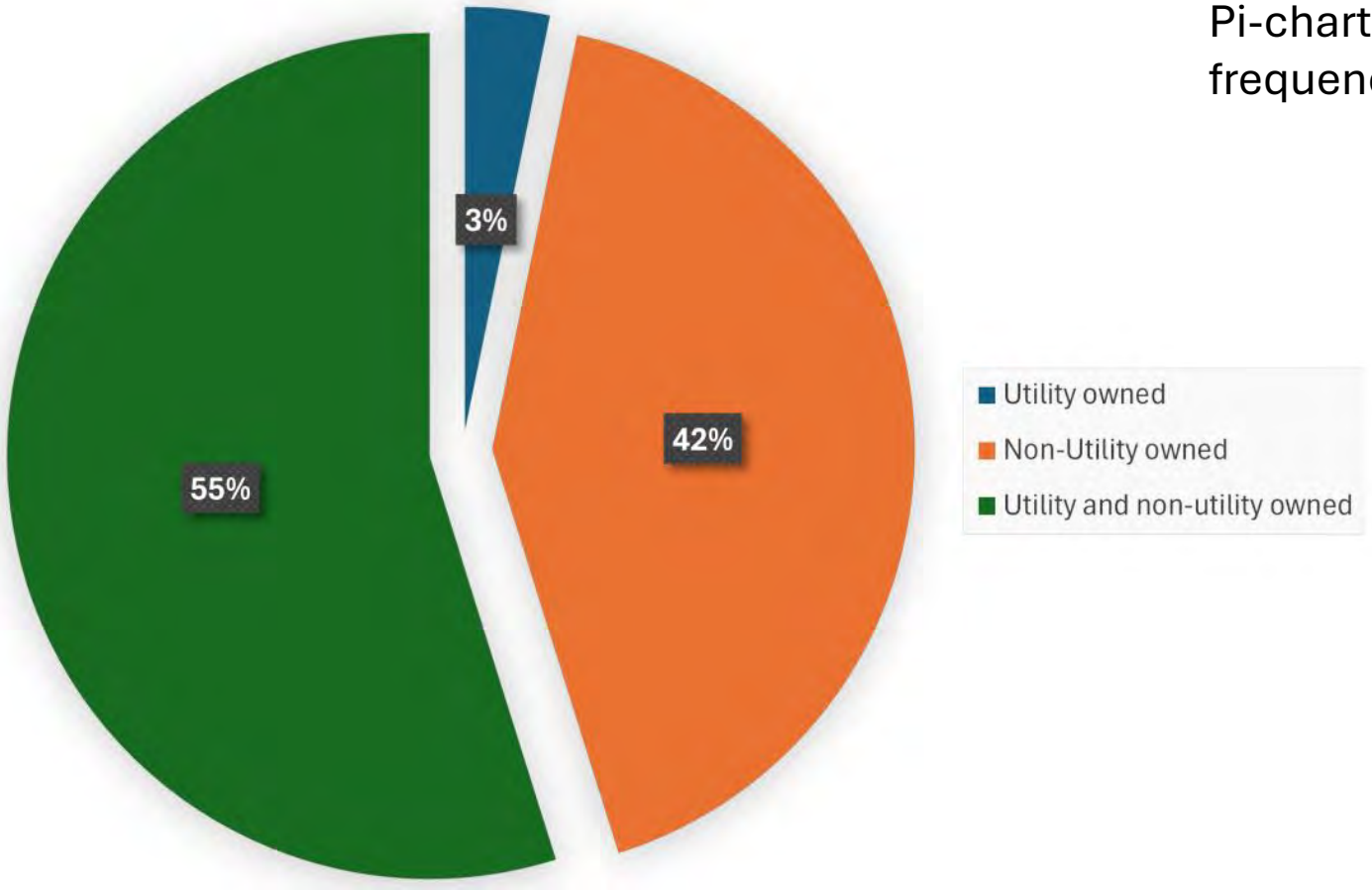
Ownership of flexibility



DER ownership

MV level

Pi-chart showing the frequency of response



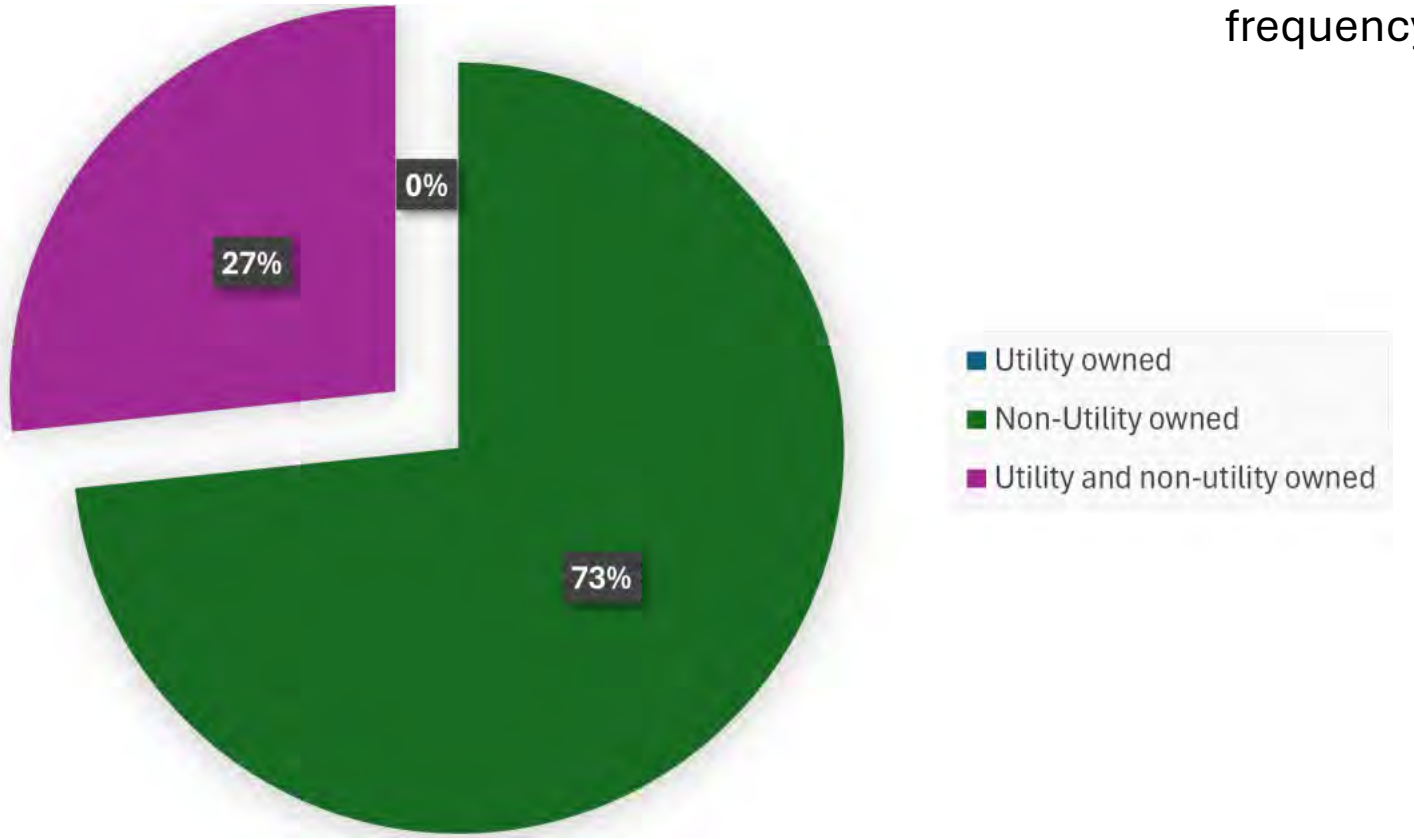
Based on survey response from 30 DSO/DNSP/DNO worldwide

Courtesy: IEEE Task force on Globally Representative Distribution Test Feeders

DER ownership

LV level

Pi-chart showing the frequency of response



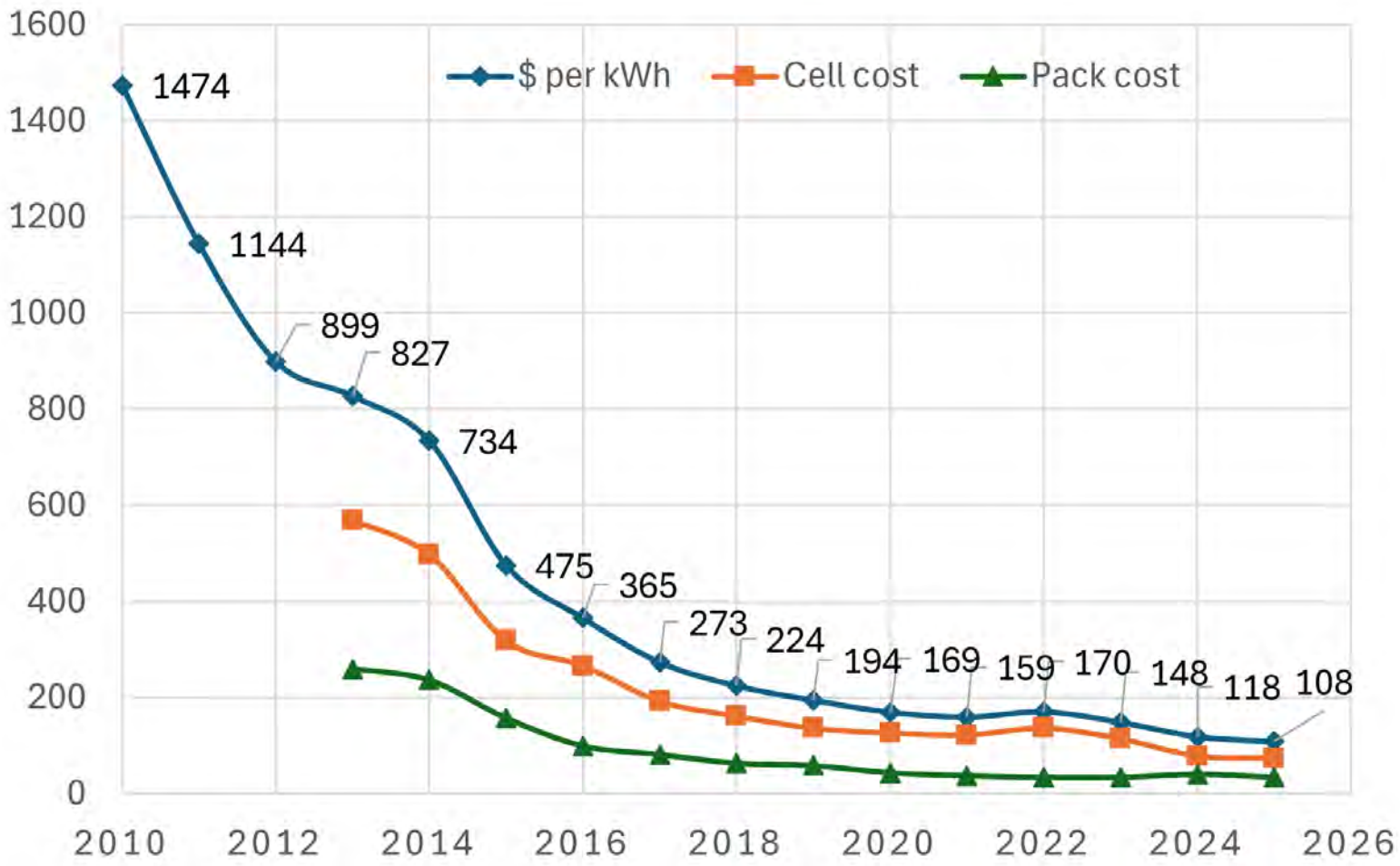
Based on survey response from 30 DSO/DNSP/DNO worldwide

Courtesy: IEEE Task force on Globally Representative Distribution Test Feeders

Stakeholder goals

For system operator	Social welfare	For flexibility owners/ operators	For passive consumers
* Ensuring power quality limits and **operational integrity** of power networks * Operational challenges mitigation/ prevention	*** Increasing share of DER** in power system * Flexibility facilitates more **efficient use of grid** without need for network upgrades * Reduced cost of consumption for responsive assts & minimize penalty for passive consumers	*** Maximize Financial incentive** in terms of reduction in cost of energy consumption is prevalent This category includes: (i) flexibility owners, (ii) flexibility service providers (FSP), (iii) aggregators	*** Minimize Financial penalty** in terms of increase in cost of energy consumption is prevalent This category includes consumers without flexibility and/or passive consumers

Batteries are becoming cheaper



Matching grid needs with available flexibility

Approach 1: Direct control of flexible assets

Applicable for

- Utility owned assets
- Contractual assets – non-utility owned

Examples of direct load control: 14a in Germany

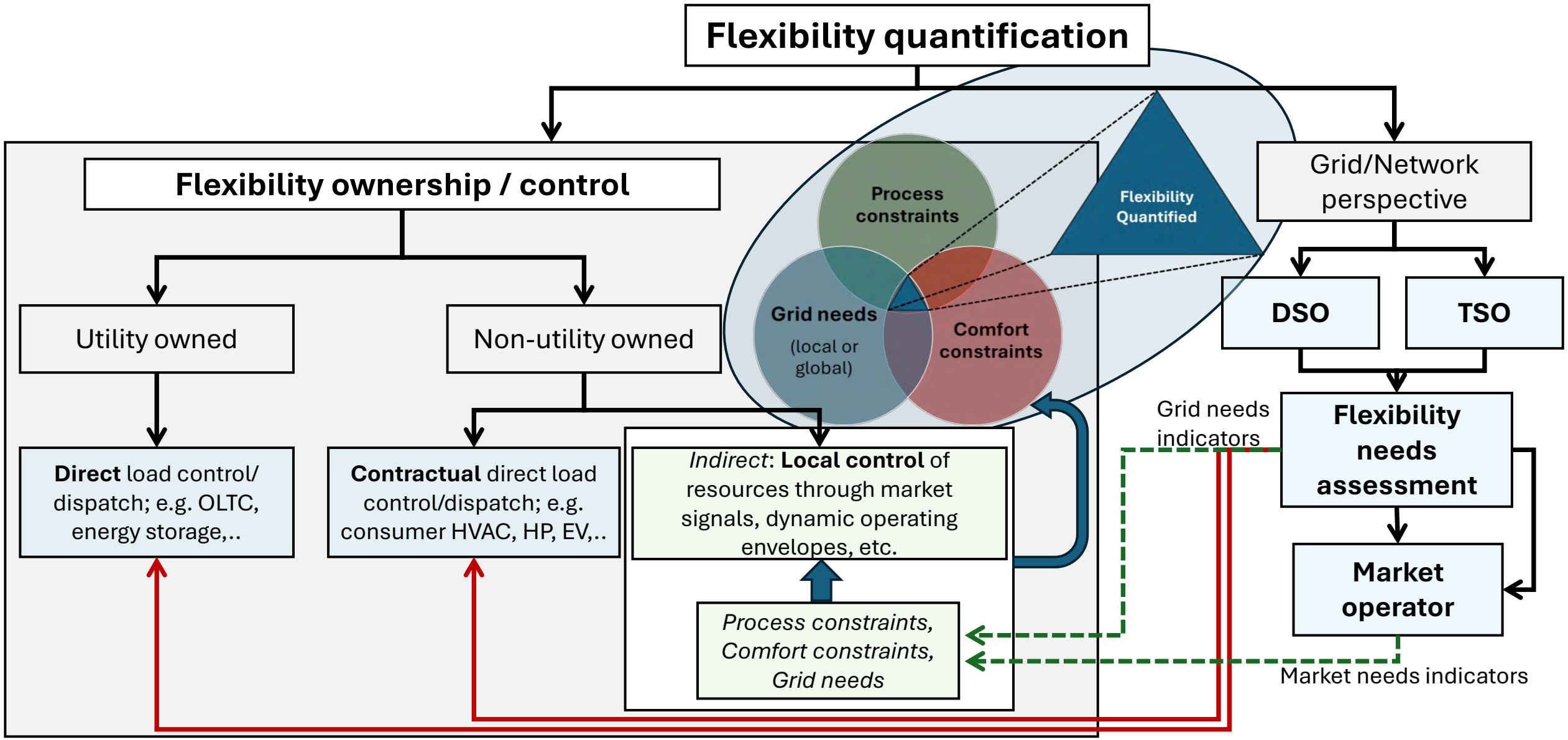
Interesting read: https://www.gridx.ai/knowledge/meter-box-14a-enwg-compliance-explained

Approach 2: Grid signal indicators (mandatory)

- Either DSO owned assets or Contractual operation
- Real time
  - Congestion indicators
  - Imbalance indicators
- Pre-set thresholds
  - Associated with proportional deterioration of power quality
    - Voltage magnitude, power factor, …
- Ex-post settlement

Approach 3: Market signal based

- Non-utility owned assets with local controllers
- Variable electricity prices
  - Time varying considering **real time** network state
  - Time varying considering **historical temporal** network state


Flexibility quantification
Flexibility ownership / control
Utility owned
Non-utility owned
Process constraints
Grid needs
(local or global)
Comfort constraints
Flexibility Quantified
Grid/Network perspective
DSO
TSO
Grid needs indicators
Flexibility needs assessment
Market operator
Market needs indicators
Direct load control/ dispatch; e.g. OLTC, energy storage,..
Contractual direct load control/dispatch; e.g. consumer HVAC, HP, EV,..
Indirect: **Local control** of resources through market signals, dynamic operating envelopes, etc.
Process constraints, Comfort constraints, Grid needs

Course acknowledgements

Kennis kent geen einde

IMPROCAP

https://www.improcap.eu/

DiGiRES

https://www.digi-res.eu/

C2-FlexIQ

https://research.kuleuven.be/portal/nl/project/3E240512

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Any questions?

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